Problem Set 1

PPHA 34410 Corporate Finance

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Part 1. Multiple Choice Questions

1) Disney's new CEO emphasizes theme parks and streaming as core growth drivers, while the company faces mixed investor sentiment and a declining stock price. Which is the best interpretation of this strategy? (3 marks)

A. Disney is shifting toward lower-risk, stable cash flow businesses B. Disney is doubling down on its strongest proven segments to stabilize investor expectations C. Disney is abandoning innovation in favor of legacy businesses D. Disney is primarily focused on cost-cutting rather than growth

2) SpaceX historically resisted going public but is now pursuing an IPO to fund space-based data centers. What is the most accurate explanation?)

C. The company needs large, immediate capital for a high-risk, capital-intensive project

3) Which of the following represents potential benefits and drawbacks of going public for SpaceX? Select all that apply. (3 marks)

A. Access to large-scale capital B. Increased short-term performance pressure C. Elimination of governance constraints D. Greater liquidity for employees E. Reduced need for strategic discipline

4) SpaceX conducts insider share sales at ~$420/share to "level-set" valuation ahead of IPO. What is the primary economic function of this? (3 marks)

A. Raising capital for operations B. Providing a credible market-based signal of valuation C. Reducing volatility in public markets D. Avoiding SEC disclosure requirements

5) Neom executives used overly optimistic assumptions and shielded leadership from bad news. What is the core failure? (3 marks)

A. Lack of capital B. Poor engineering execution C. Principal-agent problem with distorted information flow D. Insufficient demand forecasting

6) Which factors contributed to Neom's challenges? Select all that apply. (3 marks)

A. Unrealistic assumptions in financial models B. Information asymmetry between leadership and operators C. Excessively conservative planning D. Incentives to justify continued investment E. Lack of initial funding

7) Which best explains why SpaceX's ambitious projects may be more credible than Neom's? (3 marks)

A. SpaceX operates in a smaller market B. SpaceX has demonstrated execution capability and cash flow generation C. Neom is a public company D. SpaceX faces no technical challenges

8) Which conditions increase the probability that large, visionary projects succeed? Select all that apply. (3 marks)

A. Strong feedback loops B. Transparent information flow C. Unlimited capital with no accountability D. Proven intermediate milestones E. Leadership insulated from bad news

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Part 2. Short Calculation Questions (show your work for full credit)

9) Calculate how much money (future value) you will have if you invest $100,000 today. (Hint: think about "the rule of 72".) (12 marks)

a. at 10% annually for 7.2 years
b. at 4.5% annually for 16 years
c. at 8.0% annually for 9 years
d. at 9.0% annually for 8 years

10) Calculate the future value of $100,000 invested today, given the following rates and time periods, compounding annually. (12 marks)

a. r = 4%, t = 5 years
b. r = 10%, t = 5 years
c. r = 4%, t = 20 years
d. r = 10%, t = 20 years

11) Calculate the present value of $100,000 received in t number of years, given the following rates and time periods, compounding annually. (12 marks)

a. r = 4%, t = 5 years

b. r = 10%, t = 5 years
c. r = 4%, t = 20 years
d. r = 10%, t = 20 years

12) You are building a new office headquarters that is estimated to create efficiencies, saving your company $25,000,000 annually for ten years. (Assume the first savings appear one year from completion.) You will move in immediately upon completion. The building cost, which you will pay upon completion, is $140,000,000. If your company's cost of capital is 11% annually, does this appear to be a good investment? Why or why not? (5 marks)

13) Perpetuity and Annuity

a. You have the opportunity to invest in a growing perpetuity paying $100,000 per annum in year 1. The rate of growth is expected to be 6% annually. If the market rate of interest is 9% annually, how much should you be willing to pay for this perpetuity? (5 marks)

b. You have the opportunity to invest in a growing 10 year annuity paying $100,000 per annum in year 1. The growth rate is expected to be 6% annually. If the market rate of interest is 9% annually, how much should you be willing to pay for this annuity? What would you pay for the annuity if instead of 10 years, the annuity's term is 20 years? (5 marks)

14) You won your state lottery and the prize is available in several alternative payments. Which would you choose if the available market interest rate is 6% annually? Which, if the rate were 10%? Show your calculations. (16 marks)

a. $100,000,000 today
b. $18,000,000 per annum for 8 years
c. $6,000,000 per annum in perpetuity
d. $10,000,000 per annum for 20 years
e. $2,500,000 in year one, growing at 3.5% per annum in perpetuity

15) Robert Downey Jr. is reportedly earning over $100 million to return as Doctor Doom in Avengers: Doomsday and Avengers: Secret Wars. Instead of taking the $100 million in a lump sum payment in 2026, Downey is considering an option where he would earn $8 million per year for 25 years starting in 2035. What is the internal rate of return he would earn by forfeiting the immediate $100 million in favor of the deferred stream of income? (10 marks)